CSCE 411: Design and Analysis of Algorithms (Honors)

Week 7: All-Pairs Shortest Paths

Date: October 6, 2026 *Samson Zhou*

Today's questions How do we find a shortest path between *every* pair of vertices? How do we know whether an algorithm can be improved? What does this have to do with cutting a steel rod?

1 The All-Pairs Shortest Paths Problem

Let $G = (V, E, w)$ be a directed, weighted graph with $V = \{1, \ldots, n\}$. We allow negative edge weights, but assume that there are **no negative-weight cycles**. As before, $\delta(i, j)$ is the minimum weight of an i-to-j path, or ∞ if no such path exists. The empty path has weight 0.

Definition The *all-pairs shortest paths* (APSP) problem asks us to compute $\delta(i, j)$ for _____. The output is an $n \times n$ table of distances.

1.1 Choosing the right subproblems

An *internal vertex* of a path is any vertex other than its endpoints. Let

$$D^{(k)}[i, j] = \text{minimum weight of an } i\text{-to-}j \text{ path whose}$$

$$\text{internal vertices belong to } \{1, \ldots, k\}.$$

The indices i, j may be any vertices, even if they are larger than k. With no internal vertices allowed, the path is empty or uses one edge:

$$D^{(0)}[i, j] = \begin{cases} 0 & i = j, \\ w(i, j) & i \neq j \text{ and } (i, j) \in E, \\ \infty & \text{otherwise.} \end{cases}$$

If parallel edges occur, use their minimum weight.

Question 1. *When we allow vertex k as an internal vertex, what are the two possibilities for a shortest path from i to j?*

$D^{(k)}[i,j] =$ _____

2 The Floyd–Warshall Algorithm

We implement the recurrence using just one distance table. Use the convention $\infty + x = \infty$; an implementation can skip additions with an infinite operand.

```
FLOYD–WARSHALL(G, w)

D ← D^(0)
for k = 1, ..., n
   for i = 1, ..., n
      for j = 1, ..., n
         D[i, j] ← min{D[i, j], D[i, k] + D[k, j]}
return D
```

Correctness invariant After iteration k of the outer loop, $D[i,j] =$ _____ for every pair i, j.

Why the recurrence is correct.

Try one update Suppose $D[1,3] = 10$, $D[1,2] = 3$, and $D[2,3] = -2$ before phase $k = 2$. Then the update to $D[1,3]$ gives

$$D[1,3] \leftarrow _____$$

Question 2. *What is the running time of Floyd–Warshall?*

3 Is Cubic Time Optimal?

Floyd–Warshall is a $\Theta(n^3)$-time algorithm. Does that mean *every* algorithm needs cubic time?

An algorithm versus a problem Counting the three loops proves a bound for _____. It does not prove a lower bound for _____. Writing down all the answers only gives an $\Omega(n^2)$ lower bound.

The APSP conjecture For the complexity statements in this lecture, inputs are integers with magnitude bounded by a polynomial in n, so each occupies $O(\log n)$ bits. We count operations on $O(\log n)$-bit words in the standard word-RAM model.

Conjecture 1 (APSP; historical formulation)**.** *For no fixed constant $\varepsilon > 0$ is there an algorithm that solves general weighted APSP in $O(n^{3-\varepsilon})$ time.*

Such an algorithm is called **truly subcubic**. The conjecture says that the exponent 3 is optimal; it does *not* rule out every improvement over Floyd–Warshall. For example, a speedup by a logarithmic factor is compatible with it. The customary formulation also rules out bounded-error randomized algorithms.

Question 3. *Which of the following running times would refute the APSP conjecture?*

A $O(n^3/\log n)$

B $O(0.01n^3)$

C $O(n^{2.9995})$

D $O(n^{2.9}\log^{10} n)$

Why formulate a conjecture? A conjecture is a proposed statement, not a proved theorem. Researchers use reductions to show that a substantial improvement for one problem would also improve another. This connects the difficulty of many problems, even when we cannot prove unconditional lower bounds.

4 Recall: The Rod Cutting Problem

Let p_i be the selling price of one piece of length i. Given a rod of length n, cut it into positive integer lengths to maximize the sum of the prices. Let r_j be the maximum revenue for length j.

Choose the first piece If its length is i, we earn p_i and optimally cut the remaining $j - i$ inches:

$$r_0 = 0, \qquad r_j = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

The bottom-up algorithm takes $\sum_{j=1}^{n} j = \Theta(n^2)$ time and $\Theta(n)$ space.

Same example as before

i	1	2	3	4
p_i	1	5	8	9
r_i	___	___	___	____

For length 4, the best cut is _____, earning ____.

4.1 From maximizing revenue to minimizing cost

Negate the prices and revenues: $a_i = -p_i$ and $d_j = -r_j$. Then

$$d_0 = 0, \qquad d_j = \min_{1 \le i \le j} \{a_i + d_{j-i}\}.$$

We **add** the costs of two parts, and take the **minimum** over splits.

Definition: $(\min, +)$-convolution Given arrays $a[0], \ldots, a[n-1]$ and $b[0], \ldots, b[n-1]$, compute

$$c[k] = (a * b)[k] = \min_{\substack{i+j=k \\ 0 \le i,j < n}} \{a[i] + b[j]\}, \qquad 0 \le k \le 2n - 2.$$

Trying every pair (i, j) takes $\Theta(n^2)$ time. For example, if $a = (2, 0, 4)$ and $b = (1, 3, 2)$, then

$$c[2] = \underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}$$

Conjecture 2 ($(\min, +)$-convolution; historical formulation)**.** *For no fixed constant $\varepsilon > 0$ can arbitrary polynomially bounded integer arrays be convolved in $O(n^{2-\varepsilon})$ time.*

This is the **truly subquadratic** barrier. The rod-cutting recurrence has the same form, but its second array is being computed along the way.

5 And Then, Yesterday...

The APSP conjecture and the $(\min, +)$-convolution conjecture were disproved yesterday: October 5, 2026.

Josh Alman and Virginia Vassilevska Williams posted [3], *Truly Subquadratic 3SUM and Truly Subcubic APSP via Triangles in Sparse Lopsided Graphs*. For polynomially bounded integer inputs, they give:

- A deterministic $O(n^{2.9995})$-time algorithm for APSP and min-plus matrix multiplication.
- As a consequence of known reductions, a truly subquadratic algorithm for min-plus convolution.

The paper explicitly states the convolution consequence in Section 1.2. The new algorithms use algebraic techniques beyond this course.

Why is such a small improvement significant? The conjecture ruled out *every* fixed improvement in the exponent. Since $2.9995 = 3 - 0.0005$, this is enough to refute it. An asymptotic improvement does not by itself establish a practical speedup.